

PARTS LIST

Semiconductors:

- IC1 - PIC18F4431 microcontroller
- IC2-IC6 - 7805, 5V regulator
- LED1 - 5mm light-emitting diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon)

- R1 - 470-ohm
- R2-R6 - 10-kilo-ohm
- R7 - 1-kilo-ohm
- RNW1 - 10-kilo-ohm network resistor

Capacitors:

- C1-C10 - 0.1 μ F ceramic

Miscellaneous:

- S1-S13 - Push-to-on tactile switch
- BATT. - 6V, 4.7Ah rechargeable battery
- 6x3-pin berg strip male and female connectors
- 13-pin berg strip male and female connectors

Servo motor

- 6x5V, 6 kg-cm
- 1x5V, 1.5 kg-cm

holding an object. The arm has a rectangular base that can be modified to a mobile platform if required. A rotating plate rests on top of this platform. The plate is balanced on four castor balls. The plate rotates with the help of one servo fixed to the rectangular base.

The shoulder joint is mounted on the plate. This joint is made of two servos to handle a combined torque of up to 12 kg-cm to lift the entire arm. Note that only one shoulder servo was used during testing. The elbow's servo is joined to the shoulder's followed by the wrist's servo. To attain the ability to rotate an object, another servo is attached to the wrist perpendicular to the wrist joint. Finally, a gripper mechanism is attached to the rotating wrist's servo. The gripper mechanism is constructed around a smaller servo, such that both the gripper-fingers are controlled by this

single servo.

The length of the acrylic links is critical to the mechanical design. If the links are too long, the servo motors will not have enough torque to lift any load at all. If the links are too short, it will reduce the total reach of the arm and the motors might not be able to move freely without hitting each other.

Fig. 3 shows the stick diagram representing the robotic arm. A basic understanding of the torque equation and the motor rating is required to decide the lengths of the links for optimal length and final torque. A simple rule is that a higher motor torque can lead to longer arms.

Torque. The servo's capacity to lift the loads is specified in terms of torque (kg-cm/N-m/oz-in).

The torque equation is specified as follows:

$\text{Torque} = \text{Force} \times \text{Distance} \times \sin \theta$
where ' θ ' is the angle between the 'Force' acting on the shaft and 'Dis-



Fig. 4: Torque at various angles

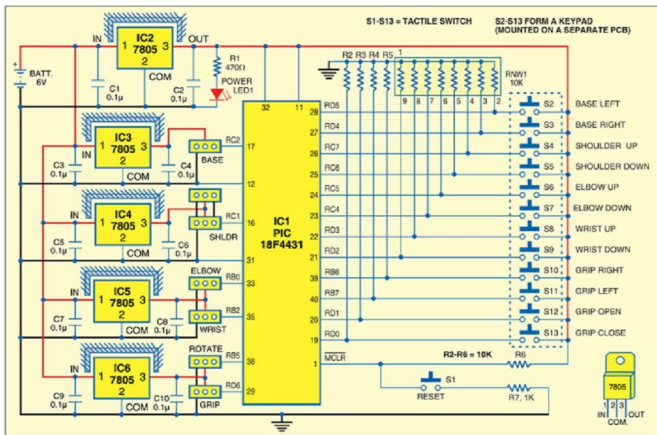


Fig. 5: Circuit of the robotic arm